

EXPERIMENT 3

Create a prototype with familiar and unfamiliar navigation elements. Evaluate ease of use with different user groups using proto.io

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AIM

To design and develop a functional, multi-screen mobile application prototype using Proto.io, focusing on user navigation and interactive design elements.

PROCEDURE

I developed a prototype using Proto.io called Campus Connect to incorporate both familiar and unfamiliar navigation elements. The aim of this prototype was to evaluate usability across different user groups and analyze how navigation styles affect user experience.

The prototype consists of four key screens:

- Login Page
- Home Page
- Events Page
- Profile Page

The main objective was to compare traditional navigation methods with gesture-based navigation techniques and evaluate which approach offers better usability.

Familiar and Unfamiliar Navigation in UI/UX Design

Familiar Navigation refers to interface patterns that users frequently encounter in mobile applications. These elements are intuitive, predictable, and reduce cognitive effort. Users can perform tasks quickly because they already understand how the interface works.

Examples:

Search bar, bottom navigation bar, login button, edit profile button, clearly labelled tabs.

Unfamiliar Navigation includes innovative or experimental interaction styles that users may not immediately recognize. These designs may improve engagement but often require adaptation and learning.

Examples:

Gesture-based navigation, hidden menus, swipe interactions, dynamic transitions.

Familiar Navigation Elements in Campus Connect

- Traditional username and password fields for login authentication.
- A clearly visible Login button placed below the input fields.
- Bottom Navigation Bar with labelled icons (Home, Events, Profile).
- Tap-based navigation for switching between screens.
- Edit Profile button positioned clearly on the profile page.
- Instant screen switching without complex animations.

These elements ensured ease of use and reduced confusion among users.

Unfamiliar Navigation Elements in Campus Connect

- Gesture-based navigation (swipe left/right to change screens).
- Hidden navigation controls without visible labels.
- Minimal text guidance, encouraging exploration.
- Dynamic transitions when moving between screens.
- Experimental interaction patterns instead of traditional tabs.

These elements were introduced to test discoverability and adaptability among different users.

User Groups for Evaluation

To evaluate usability, the prototype was tested with three user groups:

- Students (18–22 years)
- Working Professionals (25–40 years)
- Older Users (45+ years)

Participants were asked to complete tasks such as:

- Logging in

- Navigating to Events
- Checking Profile
- Returning to Home

Observations and Findings

- Students adapted quickly to both navigation styles.
- Working professionals preferred familiar navigation patterns.
- Older users experienced difficulty with gesture-based interactions.
- Familiar navigation resulted in faster task completion.
- Unfamiliar navigation required more time and exploration.

Conclusion

The evaluation showed that familiar navigation significantly improves usability, especially for diverse user groups. While unfamiliar navigation can make the interface appear modern and innovative, it may reduce accessibility and increase cognitive load.

Therefore, for applications like Campus Connect, traditional bottom navigation is recommended to ensure better user experience and accessibility.



Campus Connect

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screens











